OLLSCOIL NA hÉIREANN MÁ NUAD

THE NATIONAL UNIVERSITY OF IRELAND MAYNOOTH

MATHEMATICAL PHYSICS

# Year 3

# Semester 2

**2014 - 2015**

# Special Relativity
# MP352

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Time allowed: $1\frac{1}{2}$ hours

Answer **two** questions

All questions carry equal marks

1. Let $\mathcal{S}$ and $\mathcal{S}'$ be two inertial frames, with $\mathcal{S}'$ moving relative to $\mathcal{S}$ with a velocity of $0.6c$ in the positive $x$-direction.

   (a) An event has spacetime coordinates $(x^0, x^1, x^2, x^3)$ in $\mathcal{S}$; what are its coordinates $(x'^0, x'^1, x'^2, x'^3)$ in $\mathcal{S}'$?

   (b) Two 4-vectors $A^\mu$ and $B_\mu$ have the following components in $\mathcal{S}$:

   $$\begin{aligned}(A^0, A^1, A^2, A^3) &= (1, -1, 0, 0),\\ (B_0, B_1, B_2, B_3) &= (-1, 2, 0, 1).\end{aligned}$$

   Compute $||A||^2$ and $||B||^2$ and thus identify each vector as either timelike, spacelike or lightlike.

   (c) Find $A'^\mu$ and $B'_\mu$, i.e. the components of the two 4-vectors in $\mathcal{S}'$.

   (d) Using your results from (b) and (c), confirm that

   $$||A'||^2 = ||A||^2, \quad ||B'||^2 = ||B||^2, \quad A'^\mu B'_\mu = A^\mu B_\mu.$$

2. Let $G$ be a set and $\star$ be a binary operation on $G$.

   (a) List the four properties that $(G, \star)$ must satisfy in order to be a group.

   (b) Let $G = \{(x, m) | x > 0, m \in \mathbb{Z}\}$ and $\star$ be the operation

   $$(x, m) \star (y, n) \quad = \quad (xy, m + n).$$

   Show that $(G, \star)$ is a group.

3. Let $A$ and $B$ be two particles with rest masses $m_A$ and $m_B$ respectively, and let $X$ be a massless particle.

   (a) Prove that if either $m_A$ or $m_B$ is nonzero, then the reaction $A + B \to X$ is impossible.

   (b) Show that if $m_B > m_A > 0$, the reaction $A + X \to B$ is possible and find the resulting energy of $B$ in the rest frame of $A$.

   (c) Suppose both $m_A$ and $m_B$ are nonzero, and the reaction $A + B \to X + X$ takes place. If the total centre-of-mass energy is $E_{\rm CM}$, find the energies of $A$, $B$ and each $X$ in the centre-of-mass frame.